Tutorial-1

CSL 100 / MAL 400

Simple Program

- In this tutorial we practice the simplest C programs.

```
#include<stdio.h>
int main () {
printf("Welcome to CSL 100/ MAL 400");
return 0;
}
```

The program prints the message "Welcome to CSL 100/ MAL 400"

printf

- printf is the “voice” of the C program
  - Used to interact with the users
- printf prints its arguments in a certain format
  - Format provided by user

More printing …

```
#include <stdio.h>   int main()
{
        printf ("Hello, World! ");
        printf ("Hello \n World! \n") ;
        return 0;
}
```



\n its tell the program to stop writing on the current line and continue on the next line.

Some more printing

```
#include <stdio.h>
int main()
{
        printf ("Hello, World! \n") ;
        printf ("Hello \n World! \n") ;
        printf ("Hell\no \t World! \n") ;
        return 0;
}
```



\t its used to insert a tab space.

Data Types in C

Data type	Description	Format Specifier
int	integer	%i or %d
float	Floating point number	%f
char	Single character	%c
Char [size]	String (collection of character)	%s

Simple operation in C

operator	Description	example
+	Adds two operands	a + b
-	Subtracts the second operand from the first	a - b
*	Multiplies two operands	a * b
/	Divides the first operand by the second	a / b

Another Simple Program

- Program to add two integers (17 and 23).

```
# include <stdio.h>
int main ()
{
int a = 17;  int b = 23;  int c;
c = a + b;
printf("Result is %d", c);
return 0;
}
```

The program prints the message: **Result is 40**

Reading values from keyboard.

- scanf is used to take input from the user using the keyboard.

```
#include <stdio.h>
int main()
{
int num ;
scanf ("%d", &num) ;
printf (“No. of students is %d\n”, num) ;
return 0;
}
```



Takes input from the user and stores the value in the variable num.

Why we use & in scanf??

Reading values from keyboard.

- scanf is used to take input from the user using the keyboard.

```
#include<stdio.h>
int main()
{
char name[10] ;
char class[10];
scanf ("%s", &name) ;
scanf ("%s", &class) ;
printf (“students is %s\n from class %s”,
name,class) ;
return 0;
}
```

Two strings as input using scanf with %s, the function will read up to the first whitespace character (such as a space, tab, or newline

Reading values from keyboard.

- scanf is used to take input from the user using the keyboard.

```
#include<stdio.h>
int main()
{
char name[10] ;
char class[10];
scanf ("%s", &name) ;
Scanf (" %s", &class) ;
printf (“students is %s\n from class %s”,
name,class) ;
return 0;
}
```

The space before %s in the second scanf skips any leading whitespace, including the newline character

Class Programs

- Write a program to add two float numbers.
- Write a program to divide two float by taking input from user.

Quiz Evaluation criteria.

- Syntax: Check for and correct syntax errors like missing header files, missing semicolons, miss return or unmatched braces.
- Result: Ensure the program produces the correct and expected output.
- Structure: Verify code organization, including proper use of functions, meaningful variable names, and consistent formatting.
- Readability: Evaluate the clarity of comments and overall code readability

Quiz

- Write a program to multiply one float and decimal by taking input from user.

- Write a C program that take input from the user, user first name, last name, and favorite hobby. The program should then print a message using the input values

Ex:

Enter your first name: John
Enter your last name: Doe
Enter your favorite hobby: Reading

Hello, John Doe! It's great that you enjoy Reading.